

Report: Optimising NYC Taxi Operations

Include your visualisations, analysis, results, insights, and outcomes. Explain your methodology and approach to the tasks. Add your conclusions to the sections.

1. Data Preparation

1.1. Loading the dataset

1.1.1. Sample the data and combine the files

2. Data Cleaning

2.1. Fixing Columns

2.1.1. Fix the index

2.1.2. Combine the two airport_fee columns

2.2. Handling Missing Values

2.2.1. Find the proportion of missing values in each column

2.2.2. Handling missing values in passenger_count

2.2.3. Handle missing values in RatecodeID

2.2.4. Impute NaN in congestion_surcharge

2.3. Handling Outliers and Standardising Values

2.3.1. Check outliers in payment type, trip distance and tip amount columns

3. Exploratory Data Analysis

3.1. General EDA: Finding Patterns and Trends

3.1.1. Classify variables into categorical and numerical

3.1.2. Analyse the distribution of taxi pickups by hours, days of the week, and months

3.1.3. Filter out the zero/negative values in fares, distance and tips

3.1.4. Analyse the monthly revenue trends

3.1.5. Find the proportion of each quarter's revenue in the yearly revenue

3.1.6. Analyse and visualise the relationship between distance and fare amount

3.1.7. Analyse the relationship between fare/tips and trips/passengers

3.1.8. Analyse the distribution of different payment types

3.1.9. Load the taxi zones shapefile and display it

3.1.10. Merge the zone data with trips data

3.1.11. Find the number of trips for each zone/location ID

3.1.12. Add the number of trips for each zone to the zones dataframe

3.1.13. Plot a map of the zones showing number of trips

3.1.14. Conclude with results

3.2. Detailed EDA: Insights and Strategies

3.2.1. Identify slow routes by comparing average speeds on different routes

3.2.2. Calculate the hourly number of trips and identify the busy hours

3.2.3. Scale up the number of trips from above to find the actual number of trips

- 3.2.4. Compare hourly traffic on weekdays and weekends**
- 3.2.5. Identify the top 10 zones with high hourly pickups and drops**
- 3.2.6. Find the ratio of pickups and dropoffs in each zone**
- 3.2.7. Identify the top zones with high traffic during night hours**
- 3.2.8. Find the revenue share for nighttime and daytime hours**
- 3.2.9. For the different passenger counts, find the average fare per mile per passenger**
- 3.2.10. Find the average fare per mile by hours of the day and by days of the week**
- 3.2.11. Analyse the average fare per mile for the different vendors**
- 3.2.12. Compare the fare rates of different vendors in a distance-tiered fashion**
- 3.2.13. Analyse the tip percentages**
- 3.2.14. Analyse the trends in passenger count**

3.2.15. Analyse the variation of passenger counts across zones

3.2.16. Analyse the pickup/dropoff zones or times when extra charges are applied more frequently.

4. Conclusions

4.1. Final Insights and Recommendations

4.1.1. Recommendations to optimize routing and dispatching based on demand patterns and operational inefficiencies.

4.1.2. Suggestions on strategically positioning cabs across different zones to make best use of insights uncovered by analysing trip trends across time, days and months.

4.1.3. Propose data-driven adjustments to the pricing strategy to maximize revenue while maintaining competitive rates with other vendors.

Assumptions Made:

- Only trips with positive fare amounts and trip distances were used.
- We excluded entries where pickup or dropoff timestamps were missing or invalid.
- Pickup and dropoff zone were merged using location ID and matched using .shp file geometry.
- Night hours were defined as 11 PM to 5 AM, based on business logic.
- Tip percentage was calculated as $(\text{tip amount} / \text{fare amount}) * 100$, assuming fare includes base price but not tip.

Univariate Analysis

We examined single features like:

- **Trip Distance:** Most trips were short (~1-2 miles), indicating frequent, local travel.
- **Passenger Count:** Solo and 2-passenger trips were the most common.
- **Fare Amount:** Majority of fares were below \$20, which aligns with short trip distances.
- **Tip Amount:** Many trips had low or no tips; cash transactions may skew this data.

Visualisation: Histograms and bar plots were used to show frequency and distribution of values.

Segmented Univariate Analysis

Here, we segmented one feature based on another:

- **Trip distance by passenger count:** Found that trips with 5–6 passengers were longer on average.
- **Fare per mile per passenger:** Showed smaller groups pay more per mile, which might impact shared ride pricing.
- **Revenue by hour & day:** Weekday rush hours (7–10 AM and 5–8 PM) had higher volumes.

Visualisation: Box plots and grouped bar charts helped explain trends clearly.

Bivariate Analysis

We explored the relationship between two variables:

- **Trip distance vs. fare:** Strong positive correlation, as expected.
- **Trip duration vs. fare:** Also showed a decent correlation, but traffic seems to impact it.
- **Tip percentage vs. trip distance:** No strong trend, but higher tips seen for longer trips.

Visualisation: Scatter plots, correlation heatmaps, and bar graphs.

Geospatial Insights

- Mapped out pickup density by zone using choropleth maps.
- Found that Midtown, JFK, and Lower Manhattan were high in activity.
- Certain zones like Greenwich Village were more active at night.

Conclusions

In this analysis, I explored New York City taxi trip data to uncover patterns in demand, pricing, and operations. The goal was to understand what factors impact cab usage and revenue, and how services can be improved. Through univariate and bivariate analysis, as well as geospatial visualization, we were able to identify peak hours, busy zones, and operational inefficiencies.

Key Results:

- **Busiest Hours & Days:** Most trips occur during weekday mornings (7–10 AM) and evenings (5–8 PM). Weekend demand tends to peak later in the day.
- **Revenue Trends:** Revenue is highest in Q4, likely due to holidays. Tip percentages vary but increase slightly during late evenings and on weekends.
- **Fare and Distance:** Fare amount is closely linked with distance and duration, but fare per mile decreases slightly as trip distance increases.
- **Payment Types:** Card payments dominate. Cash usage is low and likely tied to short-distance rides.
- **Zones:** Midtown and Financial District zones see the most pickups. Late-night demand is high around nightlife areas like Times Square.
- **Passenger Trends:** Most rides have 1–2 passengers. Group rides are less common, but some zones show potential for shared or larger vehicles.

Final Insights and Recommendations

Routing & Dispatch

- Cabs should be concentrated in residential areas in the morning and commercial zones in the evening to match commuter trends.
- During night hours, reposition more cabs around high-activity zones like Greenwich Village and Chelsea.
- Identify and avoid slow routes to reduce delays and improve efficiency.

Zone-Based Positioning

- Zones with high dropoffs but low pickups (e.g., tourist zones) should get more idle cabs to reduce downtime.
- Use traffic patterns from weekdays vs. weekends to adjust cab availability in advance.

Pricing Strategy

- Introduce dynamic pricing for late-night hours or routes with historically low tip percentages.
- Consider group fare discounts for trips with 3+ passengers to encourage shared rides.
- Charge slightly higher rates for short, congested trips where trip duration is high but distance is low.